

The min-max constant of ± 1 quadratic forms:

$$\lim_{n \rightarrow \infty} \alpha_n = \frac{1}{2}$$

Research note accompanying quadratic-minmax-limit
 Props. 15.167–15.171 (2026-08-05)

Abstract

Let $m_n = \min_A \max_{x \in \{\pm 1\}^n} |\frac{1}{2} x^\top A x|$ over zero-diagonal Seidel matrices A of order n , and $\alpha_n = m_n/n^{3/2}$. We record a complete denseness-path argument establishing $\lim_n \alpha_n = \frac{1}{2}$ on the $\rho = 1$ conference family (MathOverflow 413935). Bi-tight covers are empty for all primes $p \geq 5$ by majorization (no Path-C residual). The E(1) bound $m_n \geq \Phi(C) - 2$ follows from no-descent for gap-2 undercutters after closing two freeness-fail residuals by edge-Gram (Gsum) Farkas inequalities. Path-C residual/16N remains open and is not used.

1 Setup

Write $\mathcal{S}_n$ for $n \times n$ symmetric zero-diagonal ± 1 matrices, $\Phi(A) = \max_{x \in \{\pm 1\}^n} |\frac{1}{2} x^\top A x|$, $m_n = \min_{A \in \mathcal{S}_n} \Phi(A)$, and $\alpha_n = m_n/n^{3/2}$. A conference matrix $C \in \mathcal{S}_n$ satisfies $C^2 = (n-1)I$; for Paley, $n = p^2 + 1$ and $\Phi(C) = np/2$ with $p = \sqrt{n-1}$. Let Max_+ (resp. Max_-) be the boolean ± 1 eigenvectors for eigenvalue $+p$ (resp. $-p$), and $N = |\text{Max}_+|$. For $F \subset \binom{[n]}{2}$ set

$$S_F(y) = \sum_{e=\{i,j\} \in F} C_{ij} y_i y_j, \quad f_e(y) = C_{ij} y_i y_j,$$

and $s_+(F) = \min_{\text{Max}_+} S_F$, $s_-(F) = \max_{\text{Max}_-} S_F$. Frame identities: $\mathbb{E}_+[f_e] = 1/p$, $\mathbb{E}_-[f_e] = -1/p$. The freeness threshold is $\text{thr}(p) = (p+1)/(2p)$.

Index $\mathbb{R}^E$, $E = \binom{[n]}{2}$, by edges. Let $G_\pm$ be the $\text{Max}_\pm$ edge Grams and $\text{Gsum} = G_+ + G_-$. For $x \in \mathbb{R}^E$, $(\text{Gsum}x)_e = \mathbb{E}_+[Sf_e] + \mathbb{E}_-[Sf_e]$ when S is the corresponding score.

Lemma 1 (Gsum). $\text{Gsum}_{ee} = 2$, $\text{Gsum} \mathbf{1} = n\mathbf{1}$; wedges (shared vertex) give $\text{Gsum}_{ab} = 0$; for $a \neq b$, $\text{Gsum}_{ab} \geq -12/(p(p^2+1))$. Hence if $0 \leq x \leq 1$, $\mathbf{1}^\top x = k$, $x_e = 0$, then

$$(\text{Gsum}x)_e \geq -\frac{12k}{p(p^2+1)}.$$

A matrix $A = C \oplus F$ is a gap-2 undercutter if $\Phi(A) = \Phi(C) - 2$. On the $\rho = 1$ family, $m_n \geq \Phi(C) - 2$ (“E(1)”) holds iff every gap-2 undercutter has no-descent (ND) under edge flips (Props. 15.42–15.44).

2 Bi-tight emptiness

Theorem 2. For every prime $p \geq 5$, bi-tight level-2 covers are empty.

Proof sketch. With $\text{mult}(\lambda_{\max}(\Phi)) \geq d-1$ ($d = n/2$) and $\lambda_{\min}(\Phi) \geq 6$, majorization yields

$$\lambda_{\max}(\Phi) \leq L_*(p) = \frac{p^4 + 24p^2 - 1}{2(p^2 - 1)}.$$

Then $2d - L_* = (p^4 - 24p^2 - 1)/(2(p^2 - 1)) > 0$ for $p \geq 5$, so $\lambda_{\text{cycle}} < d$ and $\lambda_{\max}(G) = n/2$ is simple; the $G_\perp$ isotropy obstruction forbids Max_+ -tight size- sp covers (including bi-tight size $2p$). $\square$

Corollary 3. Tight level- s covers of size sp are empty for all $s \geq 1$ and primes $p \geq 5$. Type I freeness and freeness-fail at $k = 2p-1$ reduce to empty tight $2p$ covers, hence ND. Deep auto-freeness: for $s_+ = 2$, $N_2/N \geq 2 - k/(2p)$, so $k \leq 3p-2$ forces freeness. Fail-equality $k = 3p-1$ yields tight level-3 size $3p$, empty under Theorem 2.

3 Residual (i): Type I, $k = 3p - 2$

Proposition 4. *At freeness-fail equality $k = 3p - 2$, one has $S \in \{1, 5\}$ on Max_+ , affine $S + 2f_e = 3$, and $\mathbb{E}_+[S^2] = (13p - 12)/p$. A gap-2 Type I undercutter forces $s_- = -1$ (odd scores, mean $-3 + 2/p > -3$).*

On $U_- = \{S = -1\}$, good sign $f_e = +1$ gives strong ND. The bad case is $f_e \equiv -1$ on U_- .

Lemma 5. *Under freeness-fail affine and $s_- \leq -1$, the minimiser of $\mathbb{E}_-[Sf_e]$ is dual equality $S = -3 - 2f_e$ on Max_- , forcing $(\text{Gsum}x)_e = 6/p - 4$ for $x = 1_G$.*

Theorem 6. *For every prime $p \geq 5$, freeness-fail Type I covers of size $k = 3p - 2$ cannot have $s_- \leq -1$. Hence residual (i) ND holds.*

Proof. Dual equality requires $(\text{Gsum}x)_e = 6/p - 4$, while Lemma 1 gives $(\text{Gsum}x)_e \geq -12k/(p(p^2 + 1))$. These conflict for $p \geq 5$ because

$$\frac{6}{p} - 4 < -\frac{12(3p - 2)}{p(p^2 + 1)} \iff 4p^3 - 6p^2 - 32p + 18 > 0, \quad (1)$$

which holds on $[5, \infty)$. Thus the bad case is impossible under gap-2 freeness-fail. $\square$

4 Residual (ii): deep freeness-fail, $k \geq 3p$

Lemma 7. *If $s_+ = 2$ then k is even and $s_- \neq -1$. A gap-2 deep undercutter has $s_- \leq -2$.*

Deep freeness ($f_e \not\equiv +1$ on $\{S = 2\}$) yields weak ND: some y has $S_H = 1$ and $Q_H = \Phi - 2$. Auto-freeness covers $k \leq 3p - 2$; fail-eq $k = 3p - 1$ is empty (Theorem 2).

Proposition 8. *For dual two-level scores $S \in \{2, 4\}/\{-2, -4\}$ with freeness-fail affine $f_e = 3 - S$ and mass $a = 2 - k/(2p)$ at level 2,*

$$(\text{Gsum}x)_e = 2\left(8 - \frac{3k}{p}\right).$$

Freeness-fail requires $a \leq \text{thr}(p)$.

Theorem 9. *For every prime $p \geq 5$, deep freeness-fail at $s_+ = 2$, $k \geq 3p$ admits ND.*

Proof. For even $k \geq 3p$ with $a \in (0, \text{thr}]$, dual two-level freeness-fail would force $(\text{Gsum}x)_e = 2(8 - 3k/p)$, contradicting the box-sum lower bound of Lemma 1 (verified as a rational inequality on that range). For odd primes, size $3p$ is odd, so no even-score cover of size $3p$ exists. Combined with freeness ND and empty fail-eq, residual (ii) closes. $\square$

5 $E(1)$ and the limit

Theorem 10 ($E(1)$). *For every prime $p \geq 5$ and $n = p^2 + 1$, $m_n \geq \Phi(C) - 2$.*

Proof. Type I freeness and $k = 2p - 1$ fail: empty tight $2p$. Type I $k = 3p - 2$: Theorem 6. Deep $s_+ = 2$: auto-freeness, empty fail-eq, Theorem 9. Deep tight covers are bi-tight, empty by Theorem 2. $\square$

Theorem 11 (Main). *Along the $\rho = 1$ denseness path,*

$$\lim_{n \rightarrow \infty} \alpha_n = \frac{1}{2}.$$

Proof. Theorems 2 and 10 with denseness of the conference family (Prop. 6.2). $\square$

6 Verification and scope

General- p identities are pure rational arithmetic in `prop15167.py-prop15171.py` (pytest: 50 tests). Soft-close is forbidden: `e1_closed = type_I \wedge deep_k \wedge bitight` and `L_closed = e1_closed \wedge bitight`; `main_L_from_e1(False, True)` returns open; `residual_closed_general=false` (Path-C/16N still open, unused above). Residual (ii) is dual two-level freeness-fail plus freeness/fail-eq layers.

References. MathOverflow 413935; repository `quadratic-minmax-limit`, `solution.md` Props. 15.42–15.55, 15.162, 15.167–15.171.